

Name: _____, Roll No. _____, Section: _____

NEDUET
CIS- 3rd Year, EEM
Assignment – 2

Total marks: 12

Date: 08/Mar/2026

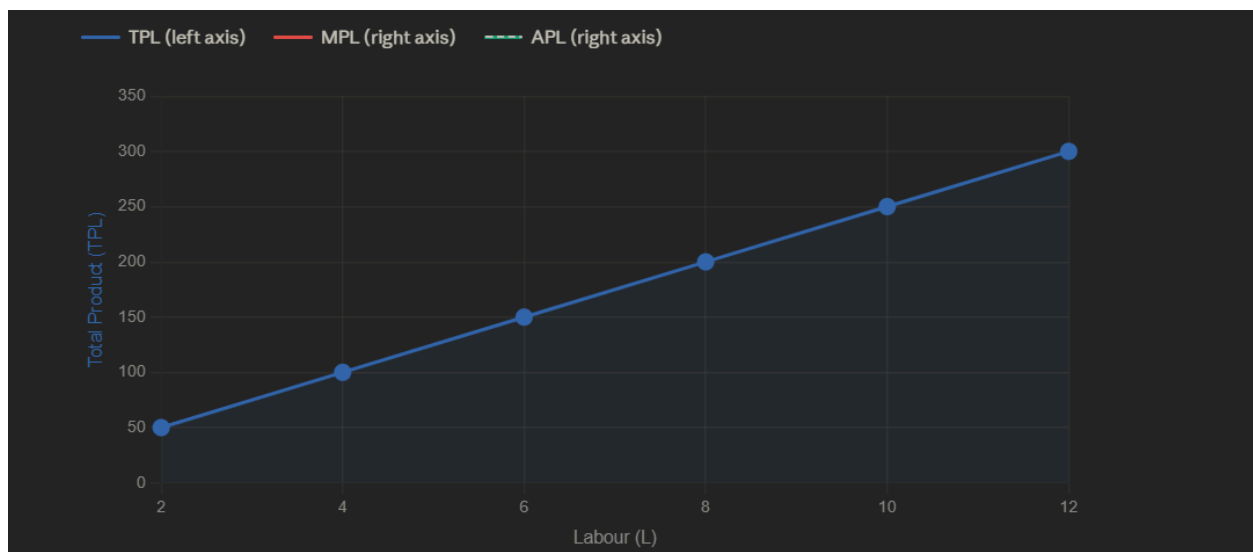
Note: Take a print and submit hand written copy on 25th March 2026. Your writing should be clearly readable. (CLO 3)

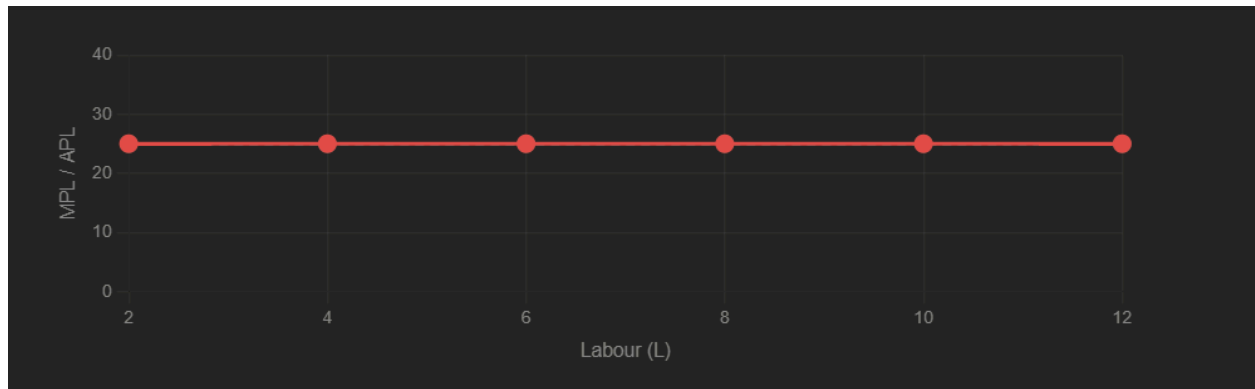
Q1. The short Run production function of a printing press is given below,

- a. **Try** to complete the following schedule assuming that the printing press/firm is experiencing constant returns of labour employment. (03)

Short Run Production Function				
Capital (K)	Labour (L)	Total Production of Labour (TP _L)	Marginal Production of Labour (MP _L)	Average Production of Labour AP _L
10	2	50	$(50-0)/(2-0)=25$	$50/2 = 25$
10	4	100	$(100-50)/(4-2)=25$	$100/4 = 25$
10	6	150	$(150-100)/(6-4) = 25$	$150/6 = 25$
10	8	200	$50/2 = 25$	$200/8 = 25$
10	10	250	25	25
10	12	300	25	25

- b. Utilize the space given below, to **reflect** the schedules of Total Production of Labour (TPL), Marginal Product of labour (MP_L) and Average Product of Labour (AP_L) on XY-Plane for the printing press experiencing constant returns of labour employment. (03)





- c. If possible, **specify** the stage of production and optimal mix of labour and capital employment of this firm. In case not possible, **share** your remarks on the stage of production and optimal mix of labour and capital employment of the firm. (03)

Answer:

Stage of production cannot be specified

Stage	Characteristics
Stage I	APL is rising; $MPL > APL$
Stage II	APL is falling, MPL is falling but positive; $MPL < APL$ — rational zone
Stage III	MPL becomes negative; TPL falls

Under constant returns to labour, **none of these stages can be identified** for this firm because:

- MPL remains fixed at 25 — it **never rises, peaks, or falls**
- APL remains fixed at 25 — it **never rises or declines**
- $MPL = APL$ at every single level of employment — there is no crossover point
- TPL grows indefinitely in a straight line — it never inflects or turns downward

The firm is essentially **stuck in a borderline zone** between Stage I and Stage II perpetually — the transition point never arrives. Classical stage analysis assumes the Law of Diminishing Marginal Returns operates, which it clearly does **not** here.

- d. Can you notice some unusual production behaviour in this firm? If yes, **share** your observations clearly and precisely. (03)

Yes, unusual behavior:

1. **MPL and APL are constant**, which is **rare in real production**. Normally:
 - Stage 1 → increasing MPL
 - Stage 2 → diminishing MPL
2. **No diminishing returns** occur, which is uncommon in the short run (fixed capital).

Observation: The printing press seems unusually efficient — adding more labor always increases output proportionally without loss of productivity.